

ROW-INCLUSION RESIDUAL DYNAMICS ON BOOLEAN RELATIONS: PREORDER TWO-CYCLES AND EXACT FIBRES

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ABSTRACT. For an $n \times n$ Boolean matrix A , replace A by the matrix that records inclusion among its row supports. The first image is always a labelled preorder, and on every preorder the update is transpose. We derive the global identity $\mathsf{T}_n^3 = \mathsf{T}_n$: nonpreorders have tail one, equivalence relations are exactly the fixed points, and all other preorders form strict transpose two-cycles. This gives every temporal fixed-set count and the dynamical zeta function. We also solve the one-step inverse problem for every target. For a preorder P , preimages are canonically bijective to labelled induced order-embedding maps from its antisymmetric quotient into the Boolean lattice $\mathcal{B}_n$. Encoding Boolean coordinates as upper sets yields an explicit inclusion–exclusion formula for $|\mathsf{T}_n^{-1}(P)|$; nonpreorder targets have empty fibre. Relational self-residuation and its preorder property are standard background and receive no contribution credit. Exact exhaustion through $n = 4$ is used only for falsification, and external release remains on hold.

1. THE LITERAL MAP AND THEOREM

Let $\mathcal{B}_n = 2^{[n]}$ be the Boolean lattice. For a Boolean matrix $A \in \{0, 1\}^{n \times n}$, write

$$R_i(A) = \{j \in [n] : A_{ij} = 1\}.$$

Define the literal self-map

$$(1) \quad \mathsf{T}_n(A)_{ij} = 1 \iff R_i(A) \subseteq R_j(A).$$

We represent a preorder P on $[n]$ by the Boolean matrix with $P_{ij} = 1$ exactly when $i \leq_P j$. Let q_n denote the number of labelled preorders on $[n]$, and let B_n denote the n th Bell number.

For a finite preorder S , let

$$J(S) = |\mathsf{Up}(S)|$$

be its number of upper sets. Given a preorder P , define $i \sim_P j$ when $i \leq_P j$ and $j \leq_P i$, and let $Q = P/\sim_P$ be the resulting quotient poset. Put

$$(2) \quad D_Q = \{(q, r) \in Q^2 : q \not\leq_Q r\}.$$

For $S \subseteq D_Q$, let Q_S be the preorder obtained by adjoining every pair in S to Q and taking reflexive–transitive closure.

Theorem 1.1 (Complete forward and inverse dynamics). *For every $n \geq 1$, the following hold.*

(i) *The image of T_n is precisely the set of labelled preorders, and*

$$(3) \quad \mathsf{T}_n(P) = P^\top$$

for every preorder P . In particular,

$$(4) \quad \mathsf{T}_n^3 = \mathsf{T}_n.$$

(ii) *Every nonpreorder has tail exactly one. The fixed points are exactly the equivalence relations; every other preorder belongs to a strict transpose two-cycle. Consequently, for every $k \geq 1$,*

$$(5) \quad |\mathsf{Fix}(\mathsf{T}_n^k)| = \begin{cases} B_n, & k \text{ odd,} \\ q_n, & k \text{ even,} \end{cases}$$

and the Artin–Mazur zeta function is

$$(6) \quad \zeta_{\mathsf{T}_n}(z) = (1 - z)^{-B_n} (1 - z^2)^{-(q_n - B_n)/2}.$$

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(iii) *A target outside the preorders has empty fibre. For every preorder P ,*

$$(7) \quad |\mathsf{T}_n^{-1}(P)| = \sum_{S \subseteq D_Q} (-1)^{|S|} J(Q_S)^n.$$

Equivalently, the fibre is canonically bijective to the set of labelled induced order-embedding maps $Q \hookrightarrow \mathcal{B}_n$; no quotient by automorphisms of Q is taken.

The operation in (1) is a row-oriented instance of relational self-residuation. Schmidt's [4, §4.4, especially Fig. 4.4.2, p. 45] right-residual convention relates a row of the numerator to the rows of the denominator that it contains; reversing that displayed orientation gives exactly the row-containment convention in (1). Relation algebra and containment preorders are standard. Principal-upper-set representations are classical [1, pp. 525–526], and induced embeddings in Boolean lattices belong to an established order-theoretic literature [3]. Our scope is the iterated finite map and its complete inverse atlas, not invention of any of those constructions.

2. COLLAPSE TO A PREORDER AND ORDER REVERSAL

Lemma 2.1 (The residual preorder). *For every Boolean matrix A , the relation $\mathsf{T}_n(A)$ is reflexive and transitive.*

Proof. Every row support contains itself, so $R_i(A) \subseteq R_i(A)$ gives reflexivity. If $R_i(A) \subseteq R_j(A)$ and $R_j(A) \subseteq R_k(A)$, then $R_i(A) \subseteq R_k(A)$, giving transitivity. $\square$

Lemma 2.2 (Principal upper sets reverse inclusion). *If P is a preorder, then $\mathsf{T}_n(P) = P^\top$.*

Proof. Row i of P is the principal upper set $\uparrow i = \{x : i \leq_P x\}$. We claim

$$\uparrow i \subseteq \uparrow j \iff j \leq_P i.$$

If the inclusion holds, reflexivity puts i in $\uparrow i$, hence in $\uparrow j$, so $j \leq_P i$. Conversely, $j \leq_P i \leq_P x$ implies $j \leq_P x$ by transitivity. Substitution in (1) proves the lemma. $\square$

Proof of theorem 1.1(i)–(ii). By lemma 2.1, the image is contained in the preorders. Conversely, if P is a preorder then so is $P^\top$, and lemma 2.2 gives

$$\mathsf{T}_n(P^\top) = P.$$

Thus every preorder occurs in the image. Applying lemma 2.2 twice after the first update gives (4).

A state is periodic only if it lies in the image, hence only if it is a preorder. On the preorder image the map is transpose, so every period is one or two. A preorder is fixed exactly when it is symmetric. A reflexive, transitive, symmetric relation is an equivalence relation, and the converse is immediate. Equivalence relations on $[n]$ are counted by B_n ; the remaining $q_n - B_n$ preorder states pair into strict cycles.

Equation (5) follows directly. Finally, using $\zeta_{\mathsf{T}_n}(z) = \exp(\sum_{k \geq 1} |\text{Fix}(\mathsf{T}_n^k)| z^k / k)$, or multiplying one factor for each cycle, yields (6). $\square$

The numbers q_n also count finite labelled topologies, a classical correspondence and enumeration problem [2]. We use this census only to state the periodic population; it is not part of the residual claim.

3. QUOTIENT-POSET GEOMETRY OF A FIBRE

Proposition 3.1 (Fibre as an induced embedding set). *Let P be a preorder and $Q = P / \sim_P$. Sending a source matrix A to its row-support map induces a bijection to labelled embedding maps*

$$(8) \quad \mathsf{T}_n^{-1}(P) \longleftrightarrow \{f : Q \hookrightarrow \mathcal{B}_n : q \leq_Q r \iff f(q) \subseteq f(r)\}.$$

No automorphism quotient is taken: the elements of Q are the distinguished equivalence classes of the fixed labelled target P .

Proof. Suppose $\mathsf{T}_n(A) = P$. By definition,

$$(9) \quad i \leq_P j \iff R_i(A) \subseteq R_j(A).$$

If $i \sim_P j$, both inclusions in (9) hold and the rows are equal. Conversely, equal rows force both relations, so the row assignment descends to an injective map $f : Q \rightarrow \mathcal{B}_n$. The two directions in (9) say exactly that f preserves and reflects order.

Conversely, from an induced order embedding f assign row support $f([i])$ to every i . Order preservation and reflection then make (9) hold, so the resulting Boolean matrix maps to P . The two constructions are inverse. $\square$

The quotient is essential. Mutual comparability in a preorder is exactly the equality forced among source rows; treating all n labels as distinct would incorrectly demand an injective embedding before this forced collapse.

4. EVERY-TARGET INCLUSION-EXCLUSION

Lemma 4.1 (Coordinate encoding). *For any finite preorder S , the number of order-preserving maps $f : S \rightarrow \mathcal{B}_n$ is $J(S)^n$.*

Proof. For coordinate $c \in [n]$, put

$$U_c = \{q \in S : c \in f(q)\}.$$

The map f is order preserving exactly when every U_c is an upper set. The n coordinates are independently chosen, giving $J(S)^n$. Conversely, the chosen upper sets recover $f(q) = \{c : q \in U_c\}$. $\square$

Proof of theorem 1.1(iii). A nonpreorder cannot be a target by lemma 2.1. Fix a preorder P and its quotient poset Q . Begin with all order-preserving maps $f : Q \rightarrow \mathcal{B}_n$. For each $(q, r) \in D_Q$, let $E_{q,r}$ be the bad event

$$f(q) \subseteq f(r).$$

Avoiding every such event is precisely order reflection, and hence, by proposition 3.1, membership in the desired fibre.

If all events indexed by $S \subseteq D_Q$ are imposed, then f must preserve the original order of Q together with every adjoined comparison in S . This is equivalent to being order preserving for the reflexive-transitive closure Q_S . By lemma 4.1, the number of such maps is $J(Q_S)^n$. Inclusion-exclusion over the bad events gives (7). $\square$

The formula remains valid when added comparisons identify elements in Q_S : upper sets depend only on the resulting preorder. Standard inclusion-exclusion and poset enumeration are used in their ordinary senses [5]; no algorithmic novelty is asserted.

5. EXACT AUDIT AND LIMITATIONS

The paper-local verifier exhausts all 2^{n^2} matrices for $n \leq 4$. For each source it independently checks preorder formation, $\mathsf{T}(P) = P^\mathsf{T}$, and $\mathsf{T}^3 = \mathsf{T}$. It then compares the formula (7) with direct fibre counts for every preorder target. Table 1 is a compact fingerprint of the frozen run.

TABLE 1. Exact theorem-interface audit. “Strict states” counts states in strict two-cycles, not cycles.

n	states	preorders	fixed	strict states	min fibre	max fibre
1	2	1	1	0	2	2
2	16	4	2	2	2	5
3	512	29	5	24	8	24
4	65,536	355	15	340	16	600

The run contains 265,050 exact assertions. Enumeration supplies no step of the proofs and says nothing about novelty beyond the tested boxes.

A second, independently implemented lane does not call the inclusion-exclusion evaluator. It enumerates labelled maps into Boolean lattices, tests order preservation and reflection, expands each map class-by-class to a source matrix, and compares the resulting set with every direct fibre through $n = 4$. It also checks all 219 labelled four-element posets in $\mathcal{B}_5$. That lane executes 13,238,845 further exact assertions; it too is only counterexample pressure.

There are also substantive limitations. Formula (7) is exact but exponential in the number of missing ordered pairs; we do not claim an efficient general evaluator. It does not simplify the still-difficult global enumeration of labelled preorders. Most importantly, the first-step self-residual construction is owned relation-algebra background. A bounded source search did not locate the precise iterated-plus-inverse conjunction, but a non-hit is not a priority or novelty certificate.

6. CONCLUSION

Row-support inclusion sends the full Boolean-relation cube onto the preorder stratum in one step, and principal upper sets turn the next update into order reversal. This produces a complete one-step-tail, one/two-cycle dynamical atlas. Quotienting mutual comparability exposes the inverse geometry: source rows are labelled induced Boolean-lattice representation maps, and upper-set coordinates make every fibre accessible to inclusion–exclusion. These statements form an internally verified theorem package; external release, priority, authorship, and submission remain on hold.

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